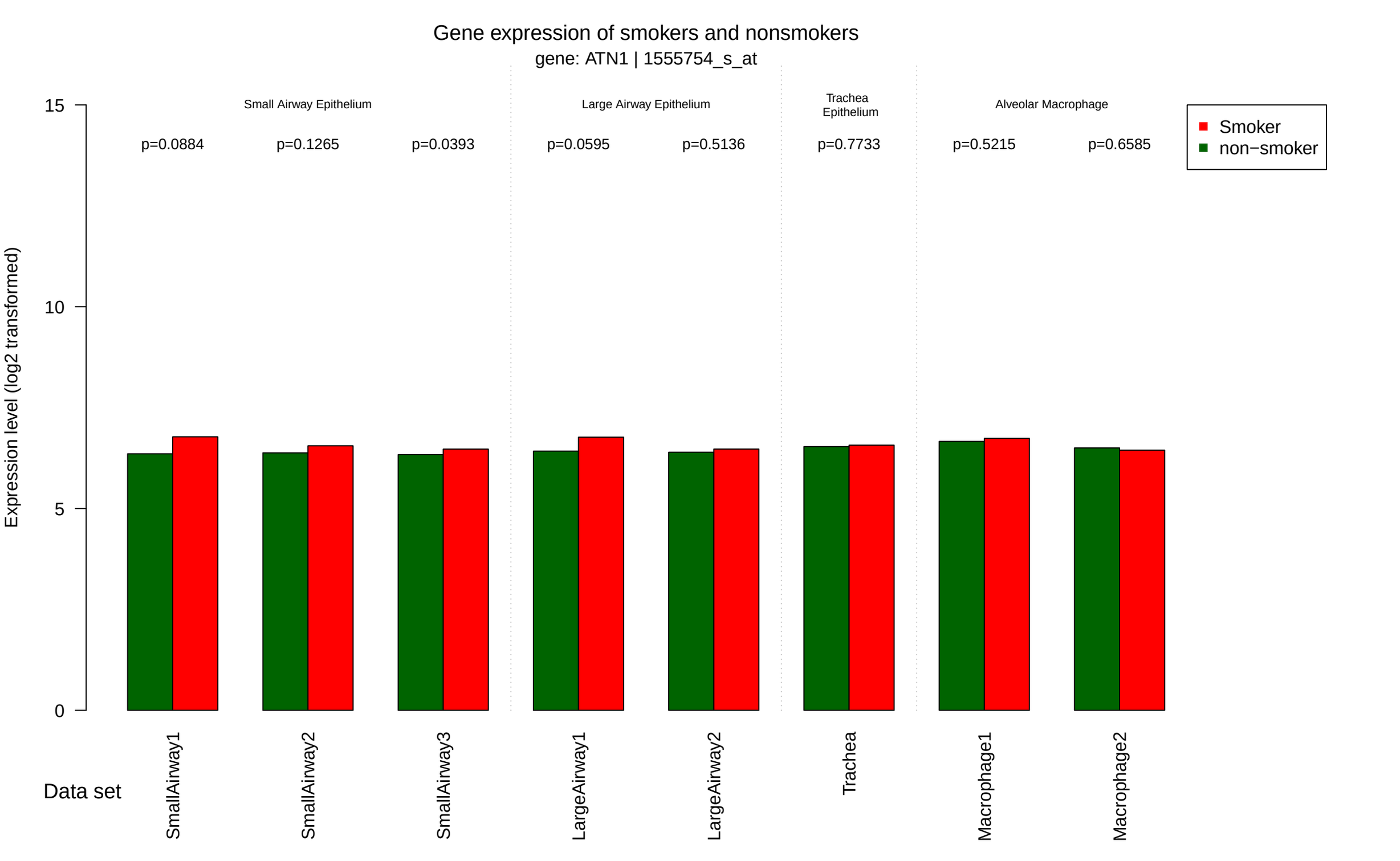
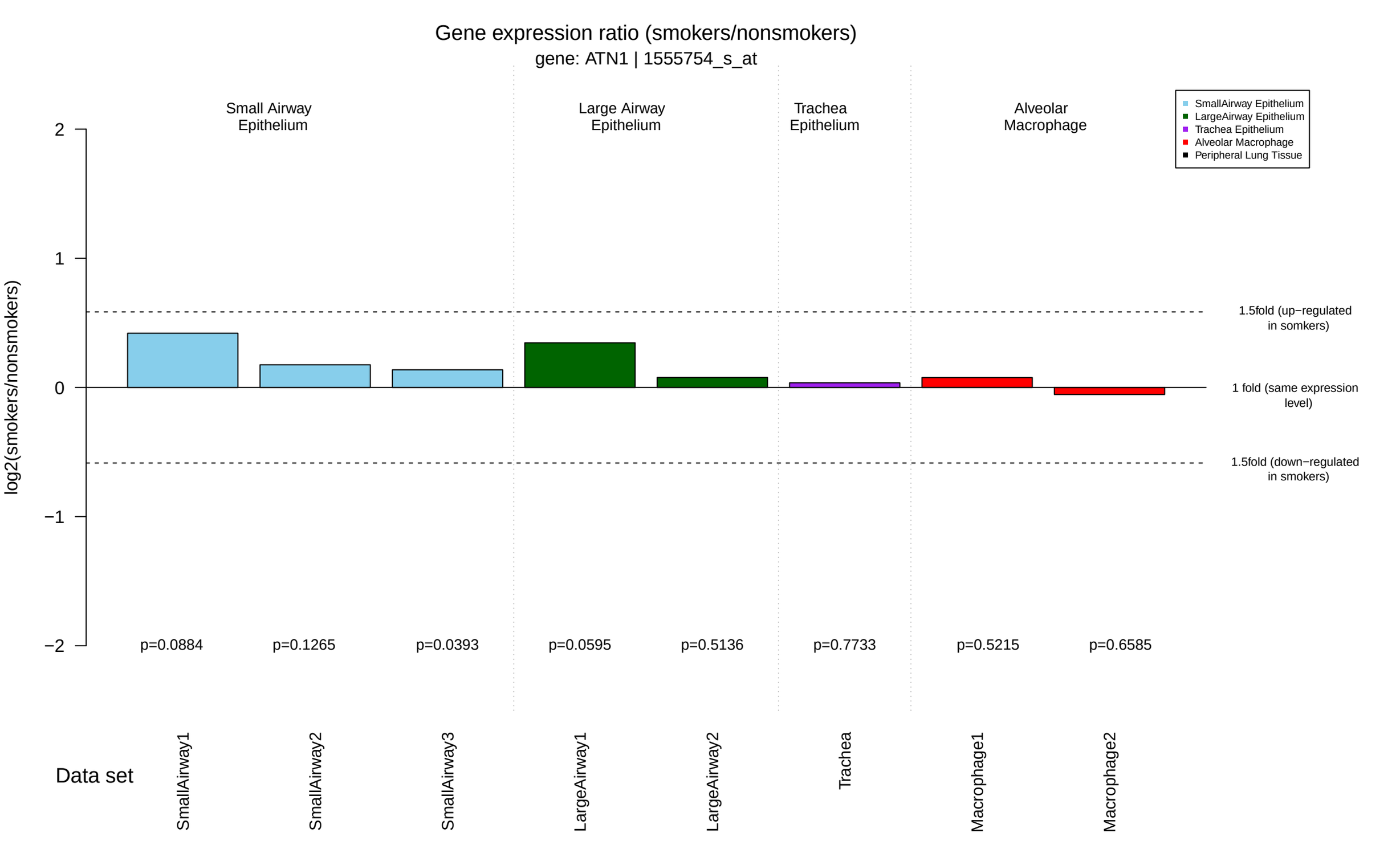
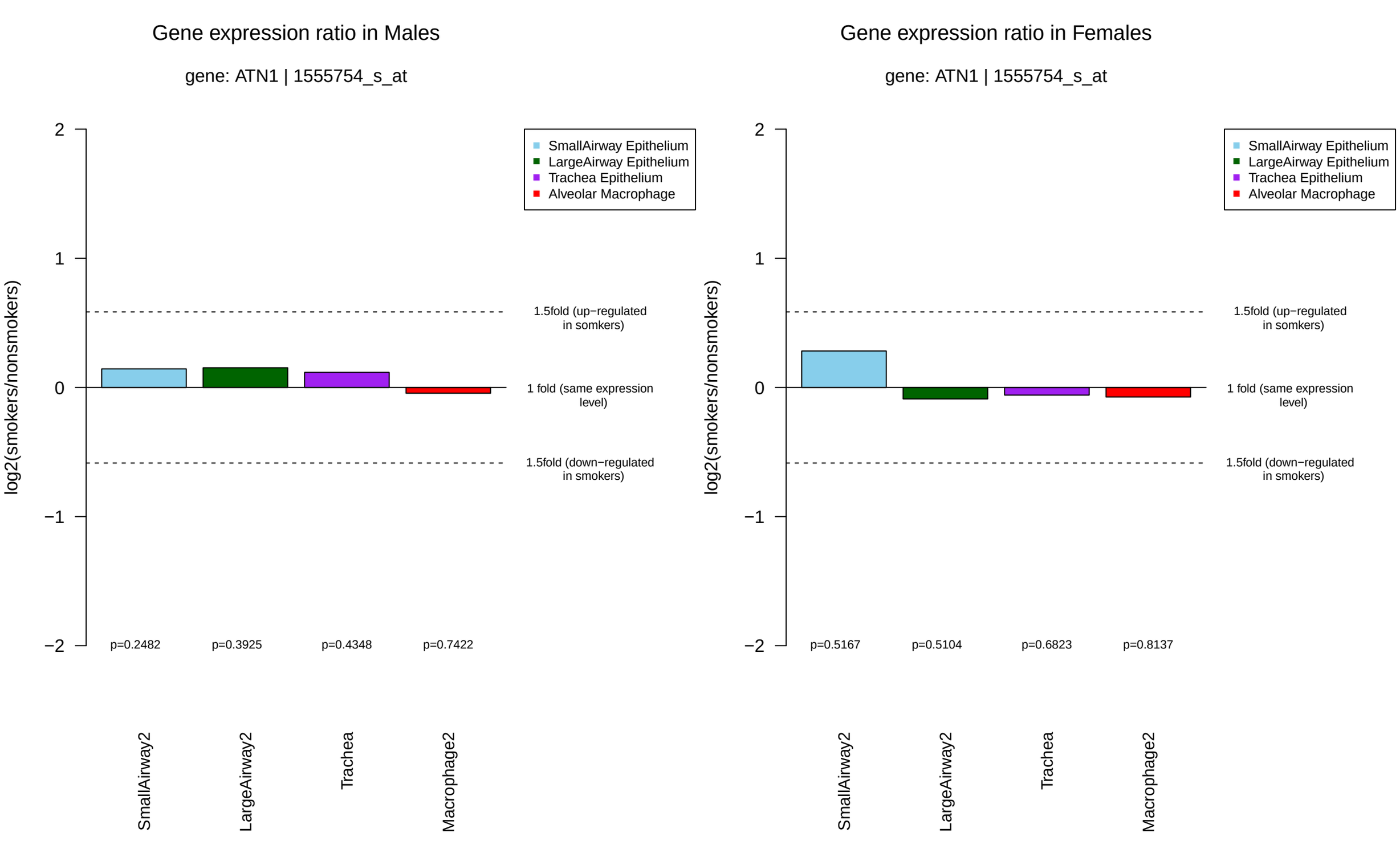
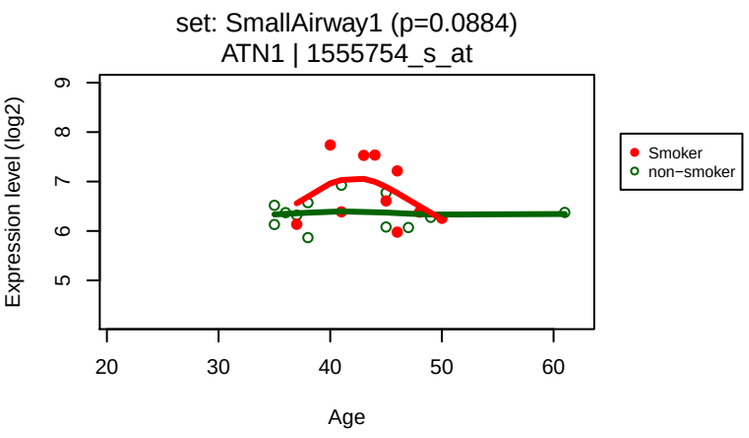
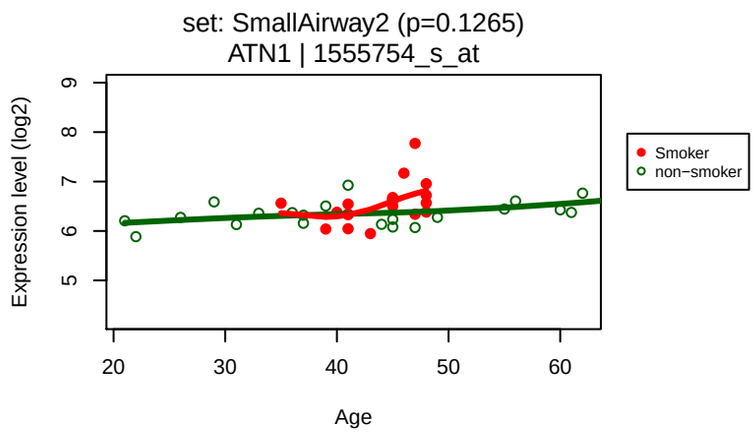
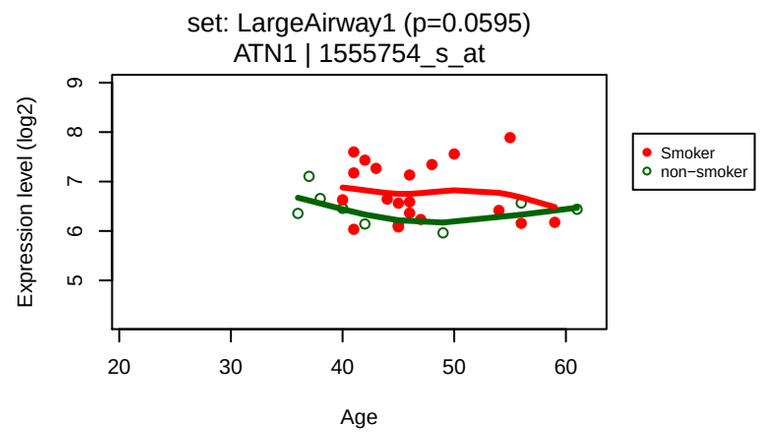

Gene expression of smokers and nonsmokers
gene: ATN1 | 1555754_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0884
p=0.1265
p=0.0393
p=0.0595
p=0.5136
p=0.7733
p=0.5215
p=0.6585
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

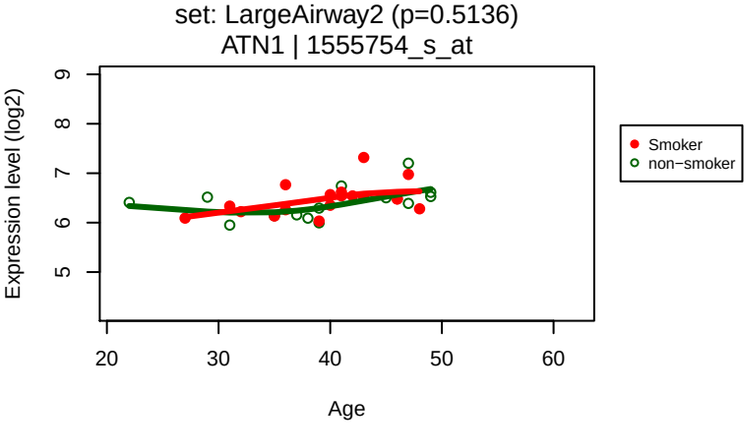

Gene expression ratio (smokers/nonsmokers)
gene: ATN1 | 1555754_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0884
p=0.1265
p=0.0393
p=0.0595
p=0.5136
p=0.7733
p=0.5215
p=0.6585
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

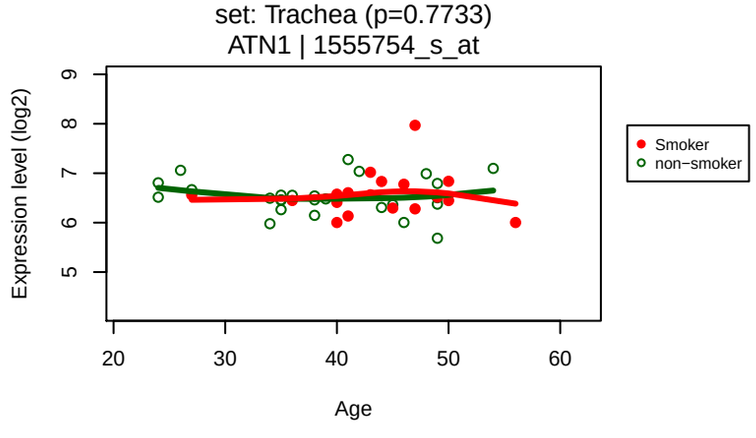

Gene expression ratio in Males
gene: ATN1 | 1555754_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2482
p=0.3925
p=0.4348
p=0.7422
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATN1 | 1555754_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5167
p=0.5104
p=0.6823
p=0.8137
SmallAirway2
LargeAirway2
Trachea
Macrophage2

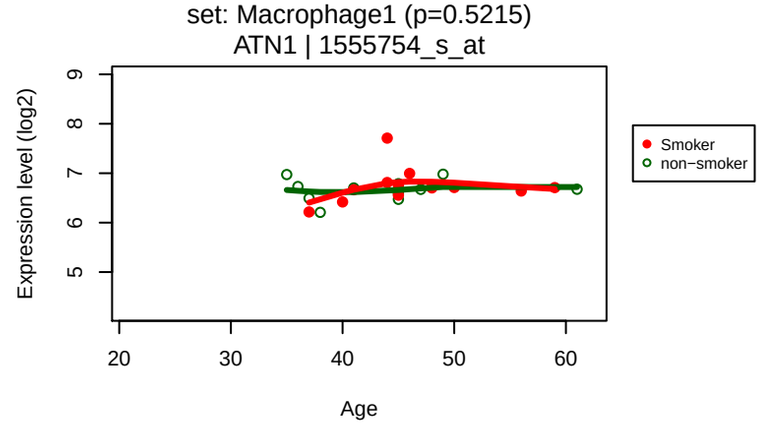

set: SmallAirway1 (p=0.0884)
ATN1 | 1555754_s_at
Expression level (log2)
Age
Smoker
non−smoker

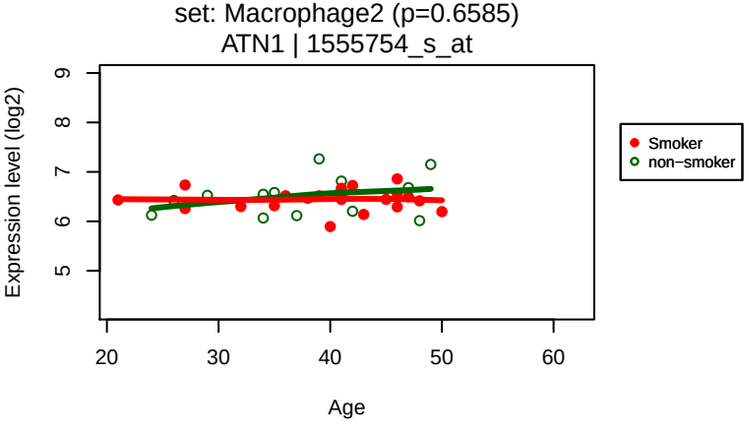

set: SmallAirway2 (p=0.1265)
ATN1 | 1555754_s_at
Expression level (log2)
Age
Smoker
non−smoker

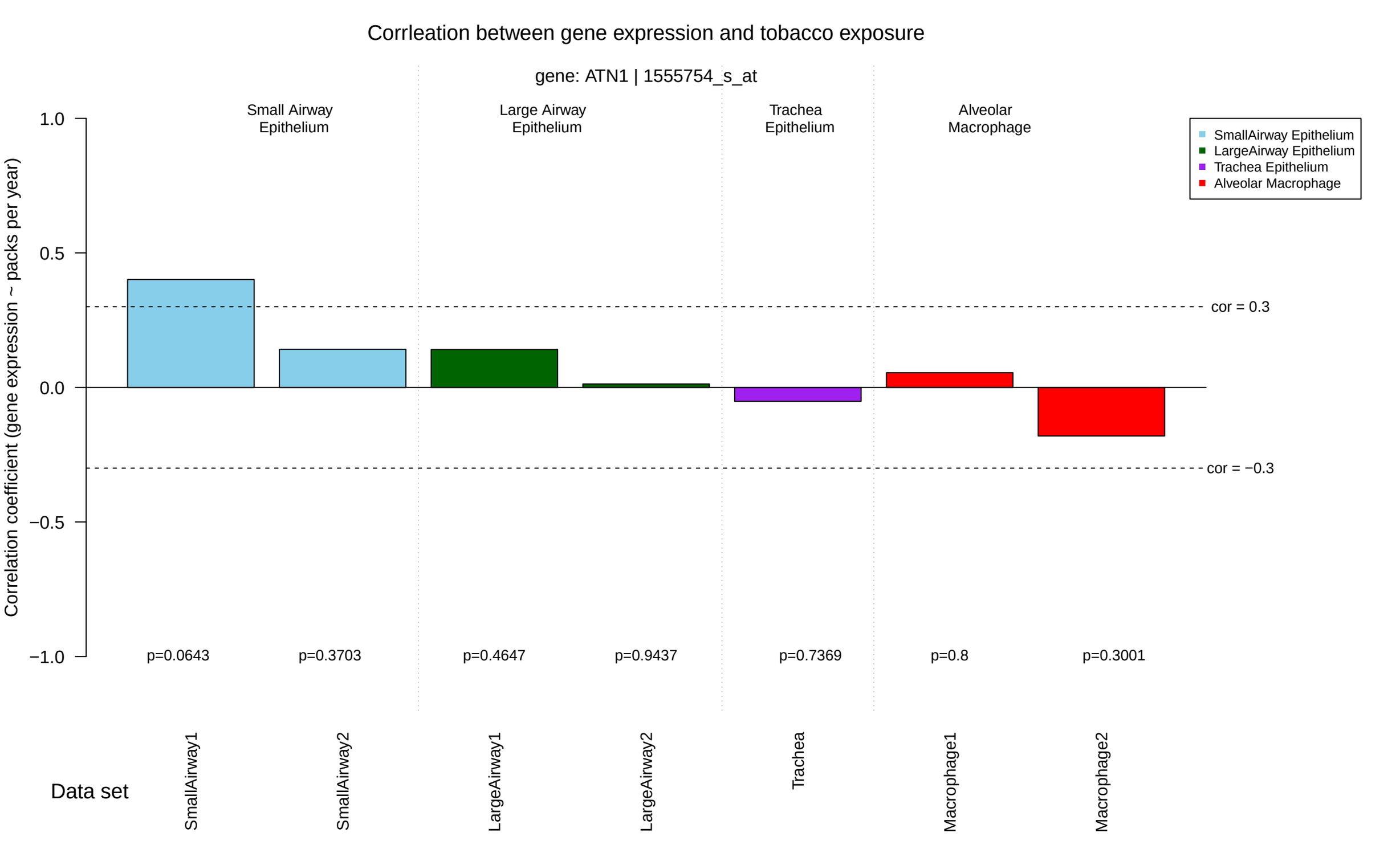

set: LargeAirway1 (p=0.0595)
ATN1 | 1555754_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5136)
ATN1 | 1555754_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7733)
ATN1 | 1555754_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5215)
ATN1 | 1555754_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6585)
ATN1 | 1555754_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATN1 | 1555754_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0643
p=0.3703
p=0.4647
p=0.9437
p=0.7369
p=0.8
p=0.3001
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2